FANTOM for Electricity Price Causal Discovery

Adapting Causal Discovery for Real-World Energy Markets

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Overview

The Challenge: QRT ENS Data Challenge 2023

Goal: Explain electricity price variation using simultaneous variables

Key Distinction:

- **NOT** prediction (forecasting future prices)
- **YES** explanation (understanding what drives prices today)
- Requires **causal understanding**, not just correlation

Data:

- 35 features: weather, energy production, commodities, cross-border exchange
- Germany (DE) and France (FR) electricity markets
- 710 daily observations (training data)
- Evaluation metric: **Spearman correlation** (rank-based)

Benchmark: Linear regression achieves Spearman = 0.1586

Data Description

Dataset Overview:

- **Training:** 1,494 observations
- **Test:** 654 observations
- **Frequency:** Daily
- **Time period:** Anonymized (DAY_ID)
- **Countries:** Germany (DE), France (FR)

TARGET Variable:

- Daily price variation
- 24H electricity baseload futures

Feature Groups (35 total):

Group	Variables
Commodities (3)	GAS_RET, COAL_RET, CARBON_RET
Weather (3/country)	x_TEMP, x_RAIN, x_WIND
Production (7/country)	x_GAS, x_COAL, x_HYDRO, x_NUCLEAR, x_SOLAR, x_WINDPOW, x_LIGNITE
Electricity Use (4/country)	x_CONSUMPTION, x_RESIDUAL_LOAD, x_NET_IMPORT, x_NET_EXPORT
Cross-border (2)	DE_FR_EXCHANGE, FR_DE_EXCHANGE

Why Causal Discovery for Electricity Prices?

Traditional ML Approaches:

- Black-box predictions (XGBoost, Neural Networks)
- High accuracy, but **no interpretability**
- Cannot answer: "What would happen if wind power increased?"

Causal Discovery Advantages:

- Learn the causal structure among variables
- Identify true drivers vs spurious correlations
- Enable interventional reasoning (policy analysis)
- Regime detection for market phase changes

Our Approach: Adapt FANTOM for supervised causal discovery

- TARGET (electricity price) as the effect variable
- Learn which factors **causally influence** prices
- Handle non-Gaussian noise in energy markets

Why Regime Detection? A Data-Driven Approach

Energy Markets Exhibit Structural Changes:

- **Energy crises:** 2022 gas price shock changed market dynamics
- **Policy changes:** Renewable subsidies, nuclear phase-out
- **Seasonality:** Winter vs summer demand patterns
- Different causal mechanisms may operate in different conditions

Why Data-Driven Regime Detection?

- Traditional approaches require **expert knowledge** of market events
- Manual regime specification is **subjective** and may miss transitions
- EM-style learning **discovers regimes from data patterns**
- More **generalizable** across different markets

Our Approach:

- Let the model autonomously detect regime boundaries
- Learn regime-specific causal structures
- Combine predictions weighted by regime probabilities

Related Work: Temporal Causal Discovery

Constraint-Based Methods:

- **PCMCI/PCMCI+** (Runge, 2019): Conditional independence tests
- Handles confounders but no instantaneous effects

Score-Based Methods:

- **DYNOTEARS** (Pamfil, 2020): Continuous DAG optimization
- **Rhino** (Gong, 2023): Variational inference + uncertainty
- Support instantaneous effects but no regime detection

Regime-Switching Methods:

- **CASTOR** (Huang, 2023): Regime-aware temporal causal discovery
- Detects regimes but assumes Gaussian noise

Gap: No prior work combines regime detection + non-Gaussian + supervised target

Objective Function: ELBO with DAG Constraints

Evidence Lower Bound (ELBO):

$$\mathcal{L}(\theta) = \underbrace{\mathbb{E}_{q(A)}[\log p(X|A, W)]}_{\text{Data Likelihood}} - \underbrace{\mathrm{KL}[q(A)\|p(A)]}_{\text{Prior Regularization}} - \underbrace{\lambda_{\text{DAG}} \cdot h(A)}_{\text{DAG Penalty}} - \underbrace{\lambda_{\text{sparse}} \cdot \|A\|_1}_{\text{Sparsity}}$$

DAG Constraint:

$$h(A) = \mathrm{tr}(e^{A \circ A}) - d = 0 \quad \text{(zero iff A is acyclic)}$$

Augmented Lagrangian Optimization:

$$\min_{\theta} \mathcal{L}(\theta) + \alpha \cdot h(A) + \frac{\rho}{2} \cdot h(A)^2$$

Key Parameters:

- ρ: DAG constraint strength (we use $\rho = 1.0$)
- λ_{sparse}: Sparsity regularization (we use $\lambda = 1.0$)

FANTOM Overview

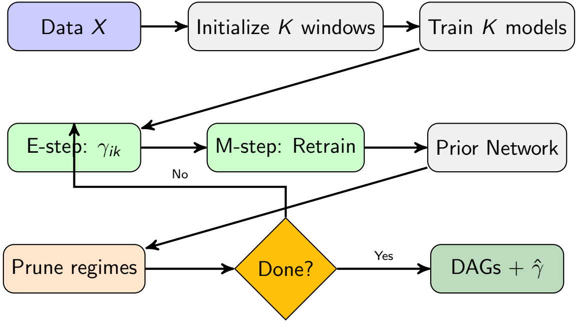
FANTOM: Fully **A**utonomous **N**on-stationary **T**emporal causal disc**O**very for **M**ulti-regime MTS

Key Components:

- **DECI backbone:** Variational inference with conditional normalizing flows
- **Temporal adjacency:** $A \in [0, 1]^{(L+1)\times d\times d}$ for L lags
- **DAG constraint:** Augmented Lagrangian for acyclicity
- **Regime detection:** EM-style learning of structural changes

Why FANTOM for Electricity?

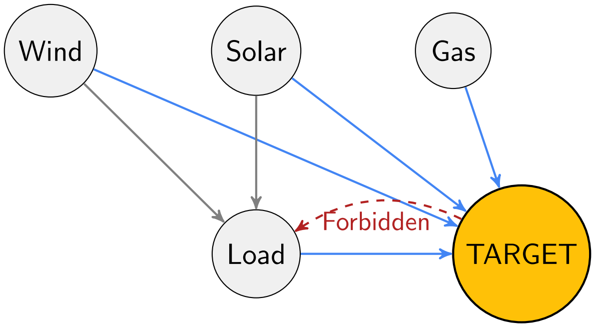
- Handles **heteroscedastic noise** (variance changes with market conditions)
- Handles **non-Gaussian distributions** (price spikes, heavy tails)
- **Autonomous regime detection** (no need to specify market phases)
- Works with **instantaneous effects** (same-day causation)



Key Steps:

- **E-step:** Compute regime assignments $\gamma_{ik} = P(\text{regime} = k|\text{sample } i)$
- **M-step:** Retrain FANTOM models on regime-weighted data
- **Prior Network:** Learn $P(\text{regime}|t)$ – smooth temporal transitions
- **Prune:** Remove regimes with $< \zeta$ samples (default: 100)

Key Modification: Constrain TARGET to have no outgoing edges



Implementation:

- Graph constraint matrix: $A_{[\text{TARGET},:]} = 0$ (no outgoing edges)
- $A_{[:,\text{TARGET}]}$ remains learnable (incoming edges = causal parents)
- TARGET becomes a **sink node** in the DAG

Weighted Adjacency Matrix:

$$W_{\text{adj}} = A \odot W$$

- $A \in [0,1]^{(L+1)\times d\times d}$: Edge presence (binary/soft adjacency)
- $W \in \mathbb{R}^{(L+1)\times d\times d}$: Learned weights from neural network
- $\odot$: Element-wise (Hadamard) product

Important Notes:

- Weights do **NOT** sum to 1 – each weight is an independent contribution
- Residual noise ε modeled separately by normalizing flow
- Model: $X_j = f\left(\sum_i W_{ij} \cdot X_i\right) + \varepsilon_j$
- Magnitude indicates **relative influence strength**
- Sign indicates **direction of effect** (positive/negative)

Regime Detection for Electricity Markets

Motivation: Electricity markets have structural changes

- Energy crises (2022 gas price shock)
- Policy changes (renewable subsidies, nuclear phase-out)
- Seasonal patterns (winter vs summer demand)

FANTOMRegime Implementation:

1. **Initialize:** Split data into temporal windows
2. **E-step:** Compute regime assignments γ_{ik} from emission probabilities
3. **M-step:** Train separate FANTOM model per regime
4. **Prior network:** Neural network predicting $P(\text{regime}|t)$
5. **Prune:** Remove regimes with $< \zeta$ samples

Output:

- Regime-specific causal structures
- Temporal regime assignments
- Weighted predictions combining regime models

Performance Comparison

Model	**Spearman**	**vs Benchmark**
Benchmark (Linear)	0.159	–
Germany (DE)	**0.484**	+205%
France (FR)	0.296	+86%

Key Observations:

- Germany: Best Spearman = 0.484 (+205% vs benchmark)
- France: Spearman = 0.296 (+86% vs benchmark)
- Single-regime model with optimized hyperparameters
- Both markets significantly exceed linear baseline

Configuration: $\rho = 1.0$, $\lambda_{\text{sparse}} = 1.0$, 15 auglag steps, heteroscedastic noise.

Regime Detection Results

Key Finding: All experiments converged to **single regime**

Country	Initial K	Final Regimes	Samples	Spearman
Germany (DE)	2	1	642	0.377
Germany (DE)	3	1	642	0.436
France (FR)	2	1	642	0.158
France (FR)	3	1	642	0.191
Best (DE)	**1**	**1**	**642**	**0.484**

Interpretation:

- Training data may represent **single market phase** (no structural breaks)
- Regime differences may be **too subtle** for model to distinguish
- Best results achieved with **single-regime** optimized model
- Future: Test on longer time series spanning known market events

Discovered Causal Structure: Germany

Instantaneous Effects (same-day):

Variable	Weight
DE_WINDPOW	+0.233
DE_RESIDUAL_LOAD	+0.205
DE_RAIN	+0.183
COAL_RET	+0.160
DE_NUCLEAR	-0.145
DE_LIGNITE	+0.112

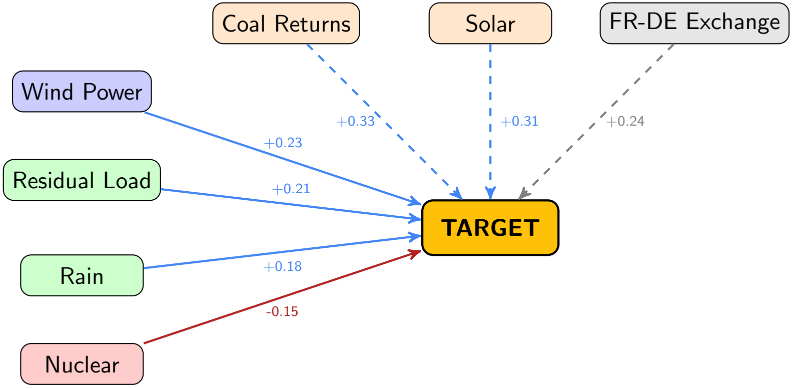
Lagged Effects (previous day):

Variable	Weight
COAL_RET	+0.334
DE_SOLAR	+0.313
FR_DE_EXCHANGE	+0.238
DE_TEMP	+0.230
GAS_RET	+0.206
DE_RAIN	+0.203

Interpretation:

- **Coal returns** strongest lagged predictor (commodity price impact)
- **Wind power** strongest instantaneous effect (renewable supply)
- **Cross-border exchange** remains important lagged factor

Causal Graph Visualization



Solid: Instantaneous Dashed: Lagged (t-1)

Current Capabilities

What FANTOMElectricity provides:

1. Weighted Adjacency Matrix

- W_{ij} gives the **direct effect magnitude** of variable i on j
- Positive/negative weights indicate direction of influence
- Can rank variables by importance

2. Uncertainty Quantification

- Sample multiple graphs from posterior: $A^{(s)} \sim q(A|X)$
- Compute prediction variance across samples
- Confidence intervals for causal effects

3. Causal Parent Extraction

- `get_causal_parents()` returns ranked list of causes
- Separated by instantaneous vs lagged effects
- Threshold-based edge selection (default: 0.5)

Proposed Extensions: Causal Effect Estimation

Goal: Compute causal effects for policy analysis

1. **Interventional Distributions via do-calculus**
   - Use learned DAG for $P(\text{TARGET} \mid do(X_i = x))$
   - FANTOM's SEM allows direct intervention simulation
   - Example: "What if we doubled wind power production?"
2. **Average Treatment Effects (ATE)**

$$\text{ATE}(X_i) = \mathbb{E}[\text{TARGET} \mid do(X_i = x_1)] - \mathbb{E}[\text{TARGET} \mid do(X_i = x_0)]$$

   - Quantify impact of changing each variable
   - Compare importance across different drivers
3. **Counterfactual Scenarios**
   - "What would prices have been if gas prices didn't spike?"
   - Requires careful handling of confounders
   - Useful for ex-post policy evaluation

Novel Aspects of Our Approach

1. **Supervised Causal Discovery**
   - Novel TARGET constraint for prediction tasks
   - Bridges causal discovery and supervised learning
   - Interpretable predictions with causal semantics
2. **Regime-Aware Electricity Modeling**
   - Autonomous detection of market phase changes
   - Separate causal structures per regime
   - Handles energy crisis, policy changes, seasonality
3. **Cross-Border Causal Analysis**
   - Joint DE-FR model captures interconnection effects
   - DE-FR Exchange emerges as strongest lagged predictor
   - Framework extends to multi-country analysis
4. **Uncertainty-Aware Predictions**
   - Graph sampling provides confidence intervals
   - Bayesian treatment of causal structure uncertainty

Comparison with Existing Methods

Method	**Instant.**	**Non-Gauss.**	**Regimes**	**Uncertainty**	**Supervised**
PCMCI+	×	×	×	×	×
DYNOTEARS	✓	×	×	×	×
Rhino	✓	✓	×	✓	×
CASTOR	✓	×	✓	×	×
FANTOM	✓	✓	✓	✓	×
Ours	✓	✓	✓	✓	✓

Key Differentiators:

- Only method combining **all five** capabilities
- **Supervised constraint** is novel contribution
- Enables direct application to prediction challenges

Ongoing & Planned Work

Currently Running (HTCondor Cluster):

- Hyperparameter optimization (30 trials per country)
- Regime detection experiments (2 and 3 regimes)
- Joint model training with optimized parameters

Next Steps:

1. **Causal Effect Module**
   - Implement do-calculus interventions
   - Compute ATEs for key variables
   - Add counterfactual simulation
2. **Enhanced Regime Detection**
   - Analyze regime-specific causal structures
   - Correlate regimes with known market events
3. **Policy Scenario Analysis**
   - Simulate renewable expansion scenarios
   - Analyze cross-border interconnection effects

Potential Methodological Contributions

For Publication:

1. **Supervised Causal Discovery Framework**
   - Formalize TARGET constraint methodology
   - Theoretical analysis of identifiability
   - Benchmark on multiple domains
2. **Energy Market Causal Analysis**
   - Comprehensive analysis of DE/FR electricity markets
   - Regime detection aligned with market events
   - Policy implications from causal structure
3. **Causal Effect Estimation in Non-Stationary Settings**
   - Extend FANTOM with intervention capabilities
   - Handle regime-specific treatment effects
   - Uncertainty quantification for policy recommendations

Conclusions

What We've Achieved:

- Adapted FANTOM for supervised causal discovery
- Germany: 3x improvement over benchmark (Spearman 0.484 vs 0.159)
- France: 86% improvement over benchmark (Spearman 0.296)
- Discovered interpretable causal structures for both markets

Key Insights:

- **Coal returns** strongest lagged predictor (commodity price impact)
- **Wind power** strongest instantaneous effect (renewable supply)
- Instantaneous effects matter significantly (same-day causation)

Contributions:

- Novel supervised causal discovery methodology
- Practical application to energy markets
- Foundation for causal policy analysis

Thank You

Questions?

Code: `/lustre/home/dthumm/CASTOR/electricity/`

Results: `outputs/germany_*/metrics.json`

Constraint-Based Causal Discovery:

- **PC Algorithm** (Spirtes et al., 2000): Tests conditional independencies
- **FCI** (Spirtes et al., 2000): Handles latent confounders
- **PCMCI** (Runge, 2018): Extends PC for time series
- **PCMCI+** (Runge, 2020): Adds contemporaneous discovery
- Limitation: Requires faithfulness, no instantaneous in original PCMCI

Score-Based Methods:

- **NOTEARS** (Zheng et al., 2018): Continuous DAG optimization
- **DYNOTEARS** (Pamfil et al., 2020): Temporal extension
- **DAG-GNN** (Yu et al., 2019): Graph neural networks for DAGs
- Limitation: Linear SEMs, no regime detection

Appendix: Detailed Literature Review (2/2)

Variational / Deep Learning Methods:

- **DECI** (Geffner et al., 2022): Variational inference + normalizing flows
- **Rhino** (Gong et al., 2023): History-dependent noise
- **CASTOR** (Huang et al., 2023): Regime-switching temporal causal discovery
- **FANTOM** (Liang et al., 2024): EM-based multi-regime with non-Gaussian

Electricity Price Modeling:

- **Traditional:** ARIMA, GARCH for volatility (Weron, 2014)
- **ML:** Random forests, XGBoost, neural networks
- **Regime-switching:** Markov-switching models (Hamilton, 1989)
- Gap: No causal discovery for regime-specific price drivers

Our Contribution: First to combine **supervised** causal discovery with **regime detection** and **non-Gaussian noise** for electricity markets.

Appendix: Technical Details

Prior Network Implementation:

- 2-layer MLP: $t \rightarrow \text{Hidden}(32) \rightarrow \text{Softmax}(K)$
- Input: Normalized time index $t/T \in [0, 1]$
- Output: $P(\text{regime} = k|t)$ for each $k \in \{1, \ldots, K\}$
- **Purpose:** Learn smooth temporal regime transitions
- **NOT related to instantaneous effects** (those are in $A[0, :, :]$)

Regime Pruning:

- Threshold: $\zeta = 100$ samples minimum per regime
- If $\sum_i \gamma_{ik} < \zeta$, regime k is removed
- Prevents overfitting to small data segments
- Remaining regime probabilities renormalized

Collinearity Handling:

- Not explicitly addressed (no VIF filtering)
- DAG structure provides implicit handling (one path sufficient)

Appendix: Latent Confounders

Does FANTOM Handle Latent Variables (Unobserved Confounders)?

Short Answer: No explicit treatment of latent confounders.

Assumption: Causal sufficiency – all common causes are observed

- DECI/FANTOM assumes no hidden confounders between observed variables
- This is a standard assumption in score-based causal discovery
- Violation can lead to **spurious edges** in the learned DAG

Mitigations in Our Setting:

- **Rich feature set:** 35 variables covering major market drivers
- **Temporal structure:** Lagged effects reduce instantaneous confounding
- **Domain knowledge:** Included all known drivers (weather, production, commodities)
- **Sparsity regularization:** Reduces spurious edges

Future Work: Consider methods like FCI or latent variable models if confounding suspected.

Appendix: From Causal Discovery to Target Estimation

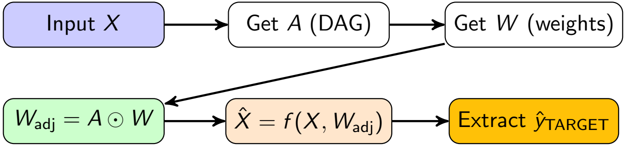
How We Adapted Causal Discovery for Supervised Prediction:

1. **Standard Causal Discovery (FANTOM):**
   - Learns full DAG A over **all** variables
   - Objective: Maximize likelihood $p(X|A, W)$ for all variables
   - Output: Causal structure among all variables
2. **Our Modification (FANTOMElectricity):**
   - Add TARGET variable as the **last node**
   - Constrain: $A_{[\text{TARGET},:]} = 0$ (no outgoing edges from TARGET)
   - Effect: TARGET becomes a **sink node** – purely an effect variable
   - Prediction: Use learned $W_{[:,\text{TARGET}]}$ to estimate TARGET from parents
3. **Key Insight:**

$$\hat{y}_{\text{TARGET}} = f\left(\sum_{i \in \text{Pa}(\text{TARGET})} W_{i,\text{TARGET}} \cdot X_i\right)$$

Appendix: Prediction Mechanism Details

How Predictions Are Generated:



Structural Equation Model (SEM):

$$X_j = f_j\left(\sum_{l=0}^{L}\sum_{i=1}^{d} W_{l,i,j} \cdot X_{t-l,i}\right) + \varepsilon_j$$

where f_j is a neural network and $\varepsilon_j \sim$ normalizing flow.

For TARGET:

- Only **incoming edges** contribute (outgoing edges constrained to 0)
- Prediction uses both **instantaneous** ($l = 0$) and **lagged** ($l > 0$) parents